We use $f_i = f(x_i)$ and define the error as

$$E(f) = \int_a^b f(x)\,dx - Q(f).$$

Closed Newton–Cotes rules

Here $h = (b-a)/(n-1)$, with $x_i = a + ih$.

Rule	Nodes (n)	Degree	$Q(f)$	$E(f)$
Trapezoidal	2	1	$\frac{h}{2}(f_0 + f_1)$	$-\frac{h^3}{12}f''(\xi)$
Simpson	3	3	$\frac{h}{3}(f_0 + 4f_1 + f_2)$	$-\frac{h^5}{90}f^{(4)}(\xi)$
Simpson's 3/8	4	3	$\frac{3h}{8}(f_0 + 3f_1 + 3f_2 + f_3)$	$-\frac{3h^5}{80}f^{(4)}(\xi)$
Boole	5	5	$\frac{2h}{45}(7f_0 + 32f_1 + 12f_2 + 32f_3 + 7f_4)$	$-\frac{8h^7}{945}f^{(6)}(\xi)$

Open Newton–Cotes rules

Here $h = (b-a)/(n+1)$, with $x_i = a + (i+1)h$.

Rule	Nodes (n)	Degree	$Q(f)$	$E(f)$
Midpoint	1	1	$2hf_0$	$\frac{h^3}{3}f''(\xi)$
Open 2-point	2	1	$\frac{3h}{2}(f_0 + f_1)$	$\frac{3h^3}{4}f''(\xi)$
Open 3-point	3	3	$\frac{4h}{3}(2f_0 - f_1 + 2f_2)$	$\frac{14h^5}{45}f^{(4)}(\xi)$
Open 4-point	4	3	$\frac{5h}{24}(11f_0 + f_1 + f_2 + 11f_3)$	$\frac{95h^5}{144}f^{(4)}(\xi)$
Open 5-point	5	5	$\frac{3h}{10}(11f_0 - 14f_1 + 26f_2 - 14f_3 + 11f_4)$	$\frac{41h^7}{140}f^{(6)}(\xi)$